AWS Cloud Security Implementation Project - EmiratesGas IAM (ISO 27001 / NIST / CIS Aligned)

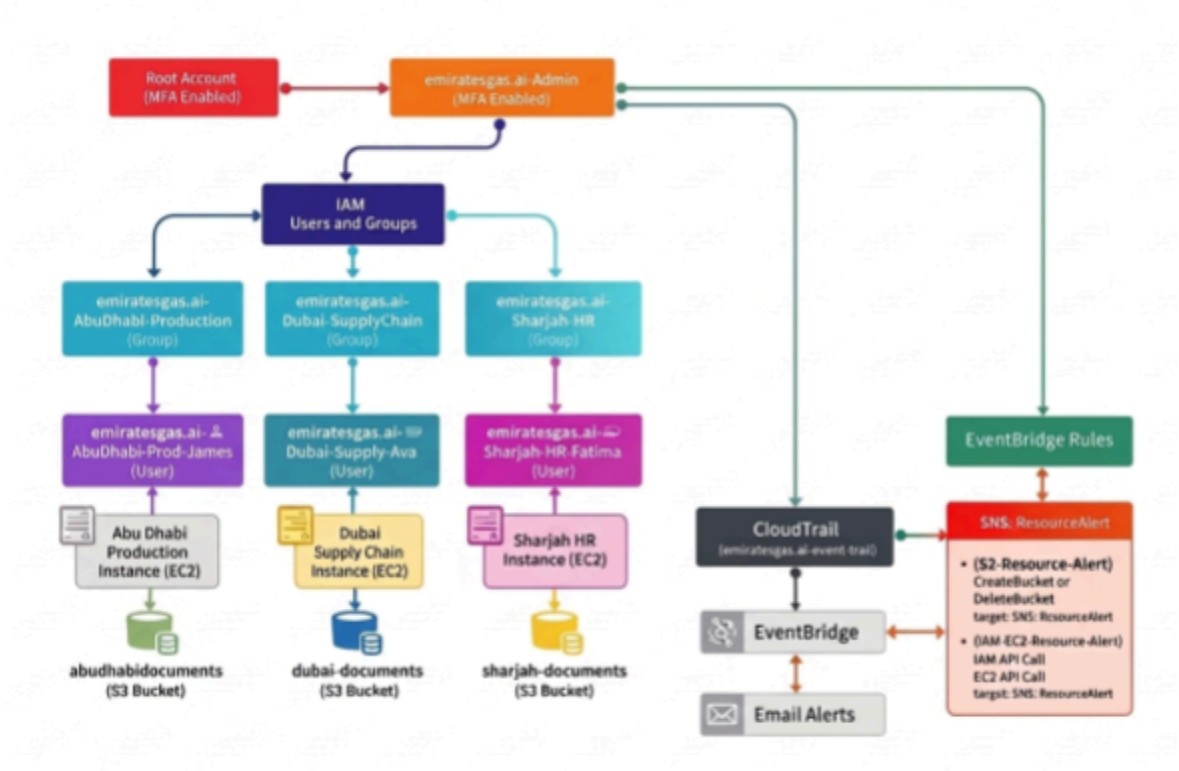
1. Overview

This project demonstrates the implementation of foundational cloud security controls aligned with **ISO/IEC 27001:2022, NIST Cybersecurity Framework (CSF), and CIS Critical Security Controls**.

The objective was to establish a **secure AWS environment** for **emiratesgas.ai** with strong identity governance, access control, logging, monitoring, and alerting mechanisms across multiple UAE locations.

The environment was designed to:

- Secure administrative access
- Enforce separation of duties
- Protect department-specific S3 resources
- Restrict EC2 administrative actions using IAM policies
- Capture and audit API activity via CloudTrail
- Generate automated alerts for sensitive S3, IAM, and EC2 actions



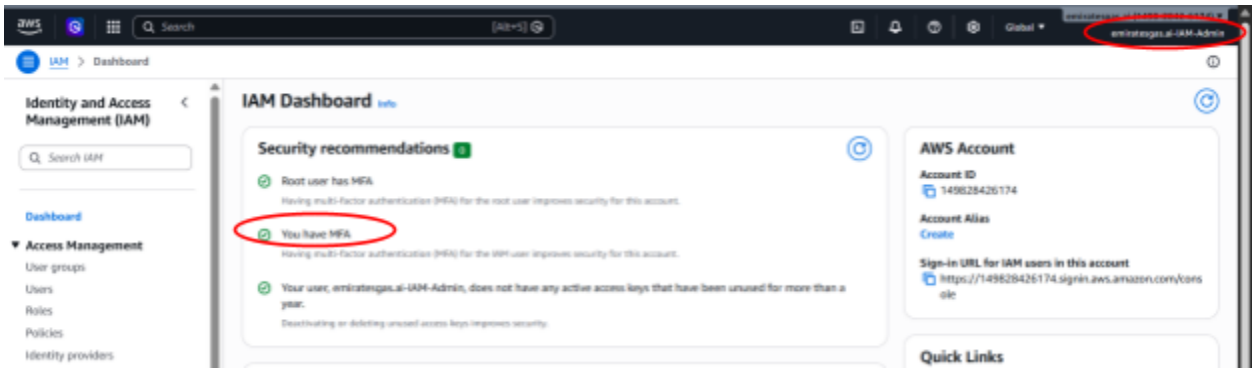
2. Governance and Account Security

ISO 27001 A.5 & A.6 | NIST CSF ID.GV | CIS Control 1

The AWS root account was secured and designated for **break-glass/emergency use only**. Multi-Factor Authentication (MFA) is recommended to further strengthen account protection.

A dedicated administrative IAM user was created to handle operational activities, ensuring:

- Accountability of administrative actions
- Reduced risk of root account misuse
- Separation between governance and operations



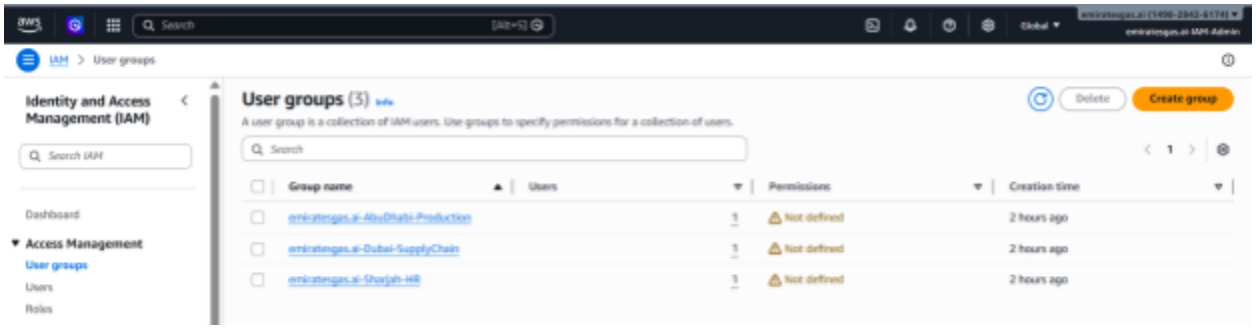
3. Identity and Access Management (IAM)

ISO 27001 A.5.15, A.8 | NIST CSF PR.AC | CIS Control 5

IAM users and groups were implemented using a **Role-Based Access Control (RBAC)** model across three business locations:

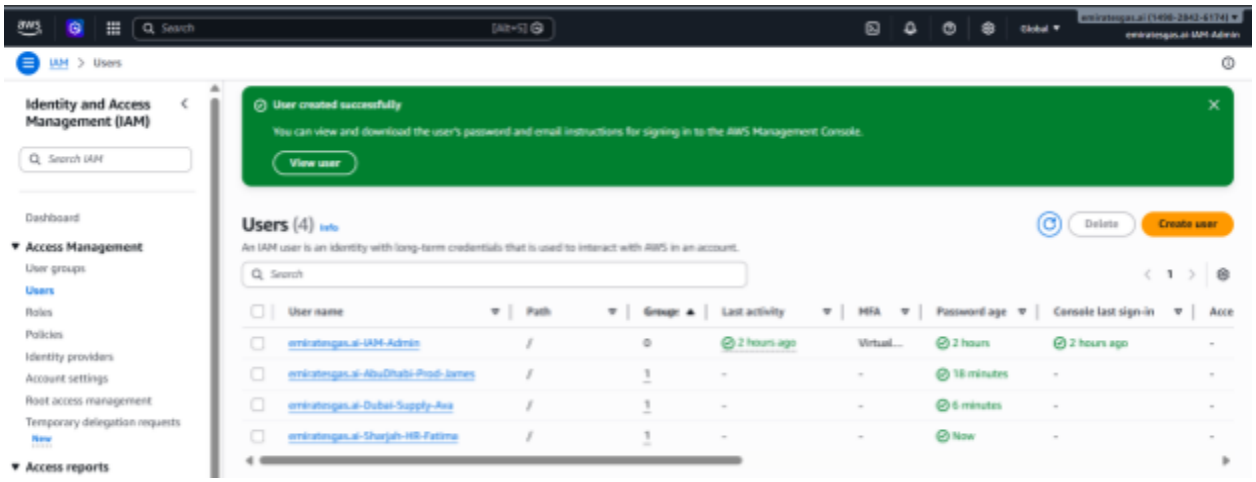
IAM Groups

- emiratesgas.ai-AbuDhabi-Production
- emiratesgas.ai-Dubai-SupplyChain
- emiratesgas.ai-Sharjah-HR



IAM Users

- emiratesgas.ai-AbuDhabi-Prod-James
- emiratesgas.ai-Dubai-Supply-Ava
- emiratesgas.ai-Sharjah-HR-Fatima



Each user was assigned to a group based on job function, ensuring:

- Least privilege access
- Separation of duties
- Controlled access to specific resources only

4. Object Storage Security (Amazon S3)

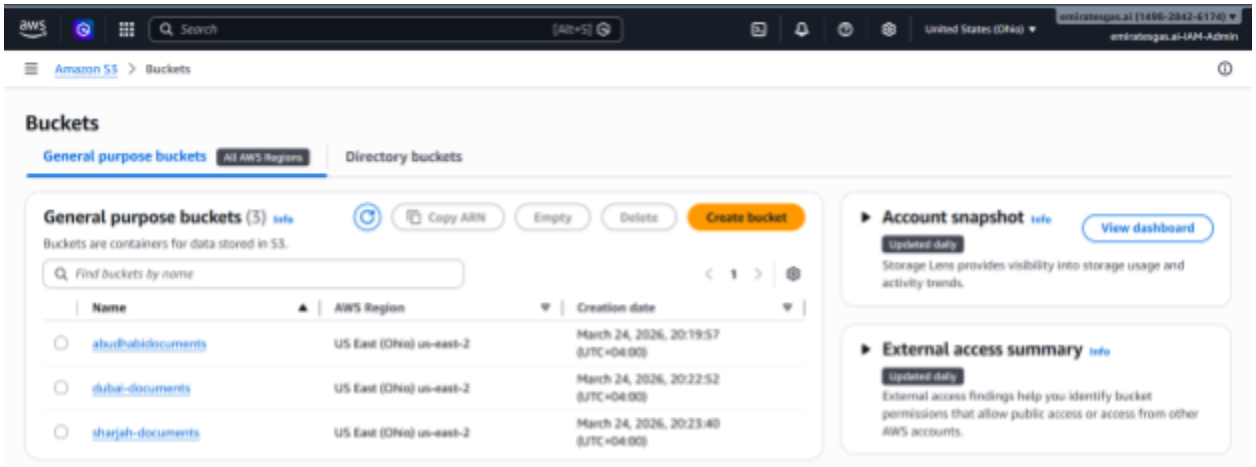
ISO 27001 A.8.2 | NIST CSF PR.DS | CIS Control 3

Three **S3 buckets** were created to isolate data per department:

- abudhabidocuments (Production)
- dubai-documents (Supply Chain)
- sharjah-documents (HR)

Security controls implemented:

- Bucket-level access restricted via IAM policies
- Access limited to assigned group users only
- Logical data isolation across departments



Access Enforcement

- Abu Dhabi user → Full access to “abudhabidocuments” only
- Dubai user → Full access to “dubai-documents” only
- Sharjah user → Full access to “sharjah-documents” only

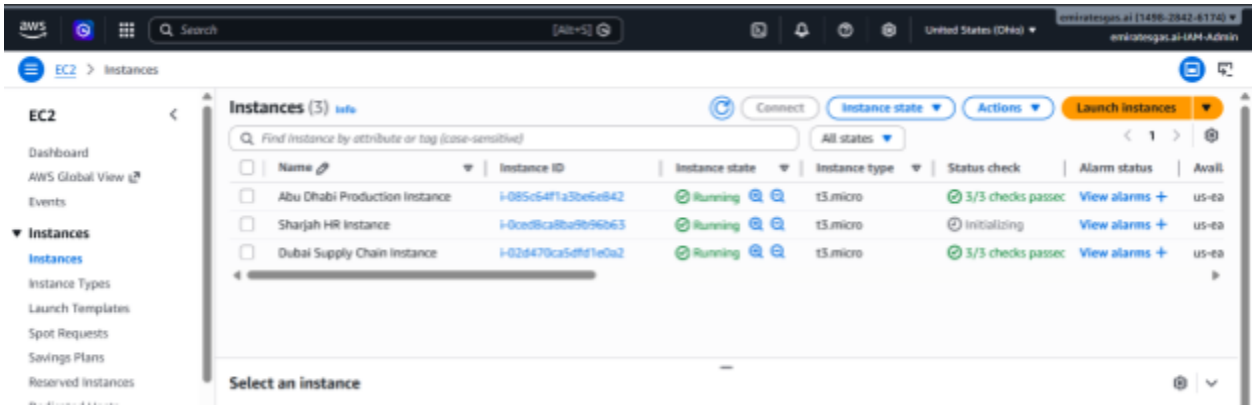
Access validation confirmed enforcement of **least privilege principles**.

5. Compute Resource Access Control (EC2)

ISO 27001 A.8.9 | NIST CSF PR.AC-4 | CIS Control 4

Three **EC2 instances** were deployed to represent departmental workloads:

- Abu Dhabi Production Instance
- Dubai Supply Chain Instance
- Sharjah HR Instance



Access control was implemented using **IAM policies with tag-based conditions**:

- Users can only start/stop instances assigned to their department
- Administrative actions are restricted across environments
- Cross-access between departments is prevented

Policy enforcement ensures:

- Controlled operational access
- Reduced risk of unauthorized instance management

6. Logging and Monitoring

ISO 27001 A.8.15 | NIST CSF DE.CM | CIS Control 8

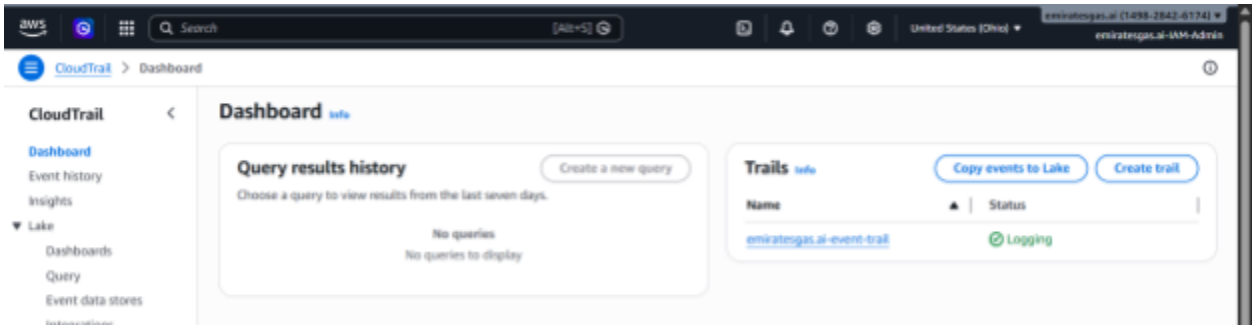
AWS CloudTrail was configured to capture and log all relevant activities.

Configuration:

1) Trail Name: `emiratesgas.ai-event-trail`
2) Logged Events:
    - S3 data events
    - EC2 and IAM API activity

Benefits:

- Full audit trail of user actions
- Support for forensic investigation
- Compliance with audit and governance requirements



7. Security Event Detection and Alerting

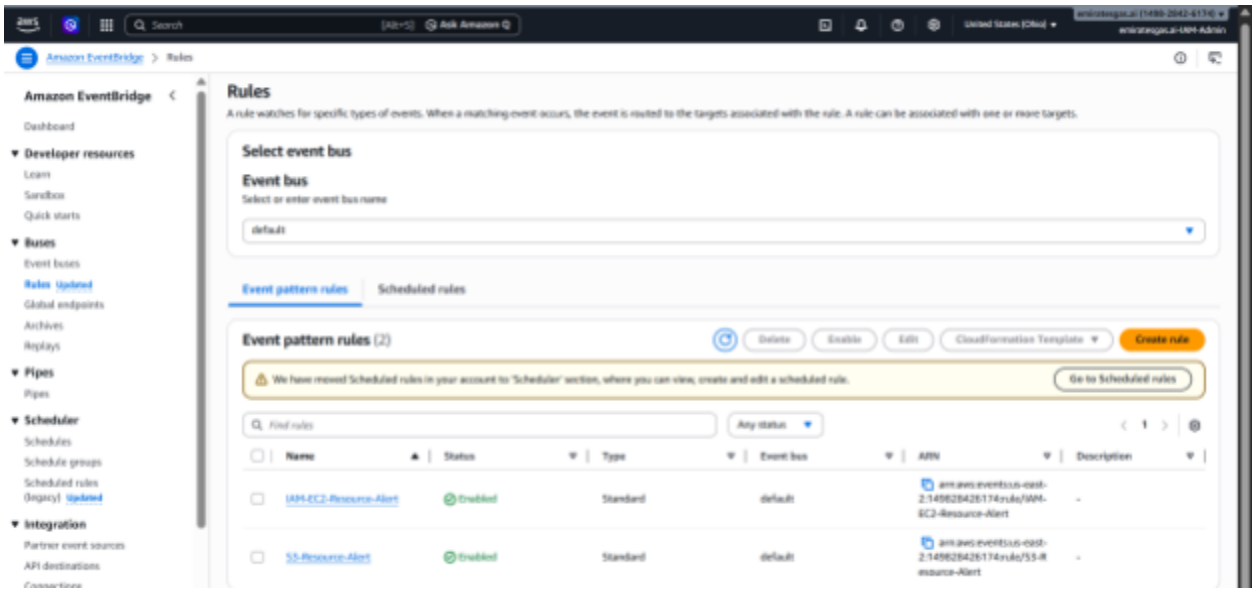
ISO 27001 A.8.16 | NIST CSF DE.AE | CIS Control 8

Event-driven monitoring was implemented using EventBridge and SNS.

EventBridge Rules

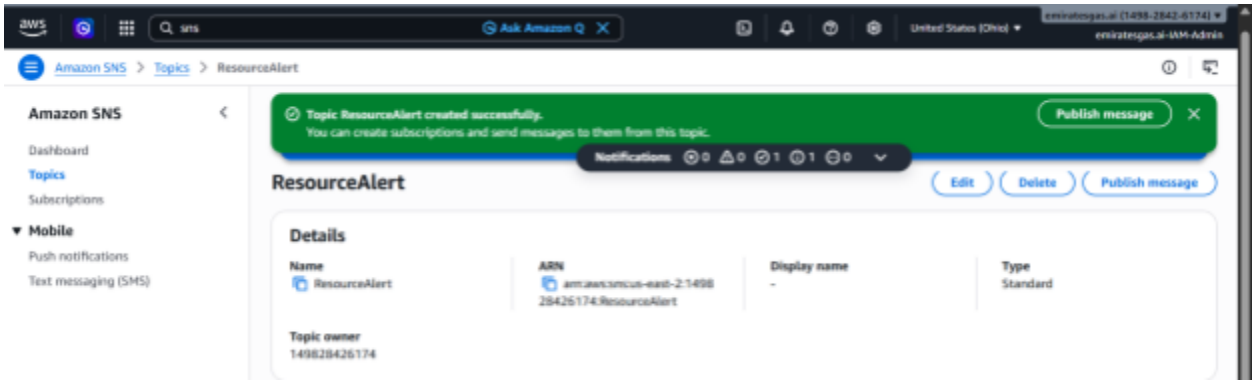
1) **S3-Resource-Alert**
   Detects:
   - CreateBucket
   - DeleteBucket

2) **IAM-EC2-Resource-Alert**
   Detects:
   - IAM API calls
   - EC2 API activity



Alerting Mechanism

- SNS Topic: `ResourceAlert`
- Protocol: Email notifications

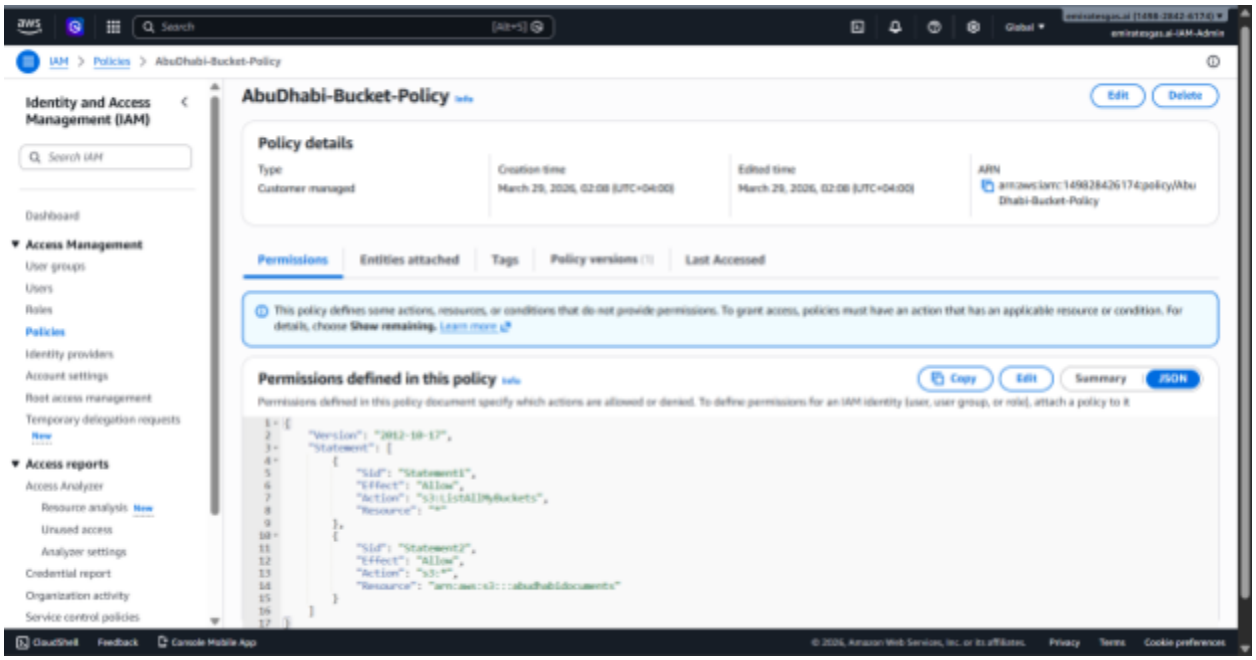


Outcome

- Near real-time detection of sensitive actions
- Immediate alerting to administrators
- Improved incident response capability

8. Custom IAM Policy (Sample)

Below is an example of the **least-privilege IAM policy** used for Abu Dhabi Production:



9. Conclusion

This project demonstrates the practical implementation of **internationally recognized cloud security frameworks** within an AWS environment.

The deployed controls align with:

- Governance (IAM, root account protection)
- Protection (S3 & EC2 access control)
- Detection (CloudTrail, EventBridge)
- Response (SNS alerting)

The solution is:

- Audit-ready
- Scalable across multiple locations
- Suitable for enterprise-grade cloud security architecture